PortableTAK Setup and Troubleshooting Guide

A complete TAK server (OpenTAKServer) that lives on a USB drive and runs on an OpenMANET node, a Windows computer, a Mac, a Linux computer or a Raspberry Pi.

1. What PortableTAK is

The drive holds the whole server: its software, its database, its users and certificates, and its settings. The computer or node it is plugged into only provides power, network and processing. Move the drive and the same server, with the same users and phones, comes up somewhere else.

- **One drive = one server.** Each drive has its own certificate authority, users and data.
- **Runs on:** OpenMANET nodes, Windows 11 computers (Intel/AMD and ARM), Apple Silicon Macs, and 64-bit Linux computers including a Raspberry Pi 4 or 5 with Raspberry Pi OS (64-bit). The drive carries software for both processor types and picks the right one automatically.
- **Moves between hosts:** on the first start on a new node or computer, the server adapts itself to that host (addresses and server certificate). Phones already set up keep working.
- **Works offline.** Internet is only needed for optional extras: Remote Access (Tailscale), online maps and aircraft/ships from the internet.
- **Dashboard:** users and phones, a **Live Map**, live aircraft and ships, video feeds you can watch in the browser, an ATAK plugin update server, data packages you can send to phones, Meshtastic radios and links to other TAK servers.
- **Dashboard layout:** on a computer screen the sections are pages in a menu on the left (the Live Map is one of them); on a phone or a narrow window they are one scrolling page.

What is on the drive

Plugged into Windows or a Mac, only a small **PORTABLETAK** volume is visible. It holds:

Start PortableTAK (Windows) / **(Mac)** / **(Linux).sh**	Runs the server on this computer.
Scan to open the dashboard (node).png	QR code for the dashboard when the drive is in a node.
Quick Start, **Quick Reference**, this guide	Documentation (PDF).

The server itself is on a second, Linux partition that Windows and macOS cannot read. If Windows offers to **format** or macOS offers to **initialize** a disk, always choose **Cancel** or **Ignore**.

2. Making a drive (PortableTAK Setup)

Drives are made with **PortableTAK Setup**, on a Windows 11 computer or a Mac (Apple Silicon). Both make the same drive.

On Windows 11

1. Unzip **PortableTAK-Windows-Setup**. Plug in a USB drive of **16 GB or more** (it will be erased).
2. Double-click **PortableTAK Setup (Windows)** and click **Yes**. The first time, it may install WSL and ask you to restart; then run it again.

3. Choose a ready-to-use drive (name and passwords now) or a blank drive, pick the USB drive from the list and type the confirmation phrase.
4. Wait about 10 minutes. The drive is built, written and verified.

On a Mac

1. One-time: install Homebrew, then in Terminal: `brew install colima docker`.
2. Plug in a USB drive of **16 GB or more**. It will be erased.
3. Double-click **PortableTAK Setup.command** (first time: right-click → Open).
4. Enter the **server name**, the **administrator password** (dashboard sign-in) and the **client password** (protects certificate packages; used for QR enrollment). Passwords: 8+ characters, 12+ recommended, and the two must be different. Typos can be retried.
5. Pick the drive from the list and type `ERASE diskN` exactly as shown. macOS asks for your Mac password.
6. Wait about 8–10 minutes: setup, writing and verification. The drive is ejected when done.

Blank drives: choose option 2 at the start to make a drive that is not set up yet (useful for preparing several drives). The first time a blank drive starts on Windows, **Start PortableTAK** asks for the server name and both passwords and sets it up on that computer. A blank drive cannot be set up from a node.

> Setup uses the Mac's internet once (map icon sets). Without internet the drive still works; the icons download later the first time the server runs with internet.

> **Never clone a set-up drive** to make more. Clones share the same keys, users and certificate authority. Always make each drive with PortableTAK Setup.

3. Using the drive on an OpenMANET node

The node needs the **PortableTAK loader** (part of the field extension, or installed once by an administrator with `portabletak-loader-install.sh`). With it, the first time a PortableTAK drive is plugged in the node installs PortableTAK from the drive by itself (about 5 minutes extra), and a drive made with a newer release updates the node the same way. That includes a drive whose Start Here files were refreshed from a newer release on a Mac or PC: the node picks up the newer package from the Start Here partition. Only packages signed with the PortableTAK release key are accepted.

1. Plug the drive into a **blue USB 3 port** on a running node. It starts by itself in 3–5 minutes, and also after every reboot or power loss.
2. Connect a phone or laptop to the node's Wi-Fi and open `https://10.41.250.10:8444` (or scan the QR code on the drive). Accept the certificate warning (Advanced → Continue).
3. Sign in as `administrator`.

Addresses on a node: `10.41.250.10` works on every node, from the node's own Wi-Fi and from anywhere on the mesh. Older CicadaTAK nodes also answer at `10.41.254.1` on their own Wi-Fi. Client packages and QR codes pick the right one.

Internet sharing: if the node's Ethernet port has internet, the node shares it with its Wi-Fi clients automatically and stops when the uplink goes away (phones then use cellular for internet).

Unplugging: press **Safe Remove** in the dashboard and wait for "safe to remove". Pulling the drive or power without this is survivable: the next start checks and repairs the drive. Avoid it when you can.

4. Using the drive on a computer (Windows, Mac, Linux, Raspberry Pi)

Windows

Requirements

- Windows 11 version 22H2 or newer (Intel/AMD or ARM). Administrator rights (Windows asks).
- Virtualization enabled (it is on most computers; WSL reports it if not).

Start

1. Plug in the drive, open the **PORTABLETAK** drive in File Explorer, double-click **Start PortableTAK (Windows)**, click **Yes**.
2. First time on a computer only (needs internet): it installs WSL (Windows Subsystem for Linux) if missing (type **Y**; then restart and start it again), installs **usbipd** (Microsoft's recommended tool that hands the USB drive to WSL), turns on WSL mirrored networking, opens the firewall for PortableTAK, and installs the small PortableTAK host. The first start of the server on a computer takes a few minutes.
3. If the drive is blank, the window asks for the server name and passwords (typos can be retried) and sets it up (5–10 minutes).
4. The window shows **this computer's address**. Phones and WinTAK use it. The dashboard opens in the browser.
5. While the server runs, the drive is attached to the PortableTAK host and disappears from File Explorer. Desktop shortcuts appear: **Stop PortableTAK**, **PortableTAK Dashboard**, **PortableTAK Status**.

Stop

Double-click **Stop PortableTAK** on the desktop. Wait for "You can remove the drive now", then unplug. On a computer the dashboard shows this reminder instead of Start/Stop buttons: only the Start and Stop shortcuts can attach and release the drive.

WinTAK on the same computer

Create a user for WinTAK and use **WinTAK Setup** in the dashboard. WinTAK connects to the computer's address shown at start.

Start automatically

Run `%LOCALAPPDATA%\PortableTAK\PortableTAK.ps1 install-helper` in PowerShell once. From then on, plugging the drive into that computer starts PortableTAK and opens the dashboard. `remove-helper` turns it off.

> **Laptop on the move:** PortableTAK uses the computer's address at start. If you change networks (for example from home Wi-Fi to a hotspot), stop and start PortableTAK so it re-binds to the new address. Phones keep their certificates. Only the address they connect to changes, so add the new address in the TAK app or use Remote Access.

Mac (Apple Silicon)

Apple Silicon Macs can run the server too, in a small Linux virtual machine. One-time preparation: install Homebrew and run `brew install qemu`.

1. Plug in the drive and double-click **Start PortableTAK (Mac)** on the PORTABLETAK volume. Enter your Mac password when asked (it gives the server access to the drive).
2. Blank drives ask for the name and passwords first. The dashboard then opens at `http://localhost:8090`, and the window shows the Mac's address for phones and WinTAK (other devices reach the dashboard at `https://<that address>:8444`). Allow incoming connections for qemu if macOS asks.
3. The PORTABLETAK volume disappears while the server runs. To stop, press Return in the window or double-click **Stop PortableTAK** on the Desktop; then eject and unplug.

Linux or a Raspberry Pi

Any 64-bit Linux computer (Intel/AMD or ARM) with systemd: Ubuntu, Debian, Raspberry Pi OS (64-bit) on a Raspberry Pi 4 or 5 with 4 GB of memory or more, and similar. Nothing needs installing first.

1. Plug in the drive. If the desktop opens the **PORTABLETAK** volume, open a terminal in it (on a Pi without a desktop: `sudo mount /dev/sda1 /mnt` and `cd /mnt`).
2. Run `sudo bash "Start PortableTAK (Linux).sh"`. The first time it installs the small PortableTAK host on this computer (about a minute, nothing of the server is stored on it), opens the TAK ports if the firewall (ufw or firewalld) is on, and stops the desktop from opening the server partition.
3. A blank drive asks for the server name and both passwords and sets itself up (5–10 minutes).
4. It shows **this computer's address** for phones and WinTAK. The dashboard is `http://localhost:8090` on this computer and `https://<address>:8444` from other devices.

From then on PortableTAK starts by itself when the drive is plugged in and when the computer boots. Before unplugging: `sudo portabletak stop` and wait for "You can remove the drive now". Other commands: `sudo portabletak start`, `status`, `uninstall` (removes it from this computer; the drive is untouched). If the computer has several networks, choose the one phones use with `sudo portabletak start --ip <address>`. A USB ADS-B radio (RTL-SDR stick) works here too: PortableTAK keeps Linux's TV-tuner drivers away from it (those drivers would otherwise claim the stick), so the computer can no longer use an RTL2832 stick as a TV tuner.

> The computer should not run another TAK server or another program on the same ports (8089, 8443, 8444, 8446, 8554). If the computer already runs Tailscale itself, PortableTAK's own Remote Access may conflict with it.

5. Using the dashboard

The dashboard arranges itself for the screen it is on:

Screen	Layout
Computer (window wider than about 1100 pixels)	A menu on the left with the server's name, **Dashboard** (status cards and Controls), **Live Map** and every section with its icon, and **Sign out** at the bottom. Click an entry to open it as a page; the server name and RUNNING badge stay at the top. The dashboard remembers the page you had open, and a page can be bookmarked (for example `https://<server>:8444/#users`). On the **Video Feeds** page every live feed plays right on the page.
Phone, tablet or narrow window	One scrolling page: the status cards, then each section, which opens when you tap its title. 🗺️ **Live Map** at the top opens the map, and **Watch** plays a video feed.

Resizing the window switches between the two. Both have the same sections and work the same way.

6. Adding phones and devices

1. **Users** → **Create User**: one user per person. In each TAK app, give every device its own **callsign**.
2. Under **Manage User / Download Setup Package** choose one:

Device	Use	Notes
ATAK (Android)	**ATAK Package**	Import → Local SD. Contains every server address (Wi-Fi, mesh, Tailscale). Wi-Fi and Tailscale entries are enabled. The others are present but off.
iTAK (iPhone)	**iTAK Package** + **iTAK Remote Package**	iTAK only uses the first server in a package. Import both for Wi-Fi and cellular.
WinTAK	**WinTAK Setup**	Import in WinTAK.
ATAK / iTAK by QR	**ATAK QR Setup** / **iTAK QR Setup**	The device must trust the server first. For ATAK, import the **ATAK Trust Package (for QR)** from Client Setup once per phone, then scan. A QR code works for 3 uses. In the QR window, pick the network the phone is on (**Phone is on:** Wi-Fi, LAN or Tailscale) if it is not already selected. **iPhone, once:** open the dashboard in Safari on the iPhone → Client Setup → **iPhone Trust Profile (for QR)** → Allow → Settings → Profile Downloaded → Install; then Settings → General → About → Certificate Trust Settings → switch on the PortableTAK CA. Without this, iTAK's QR sign-in fails with "error connecting to the server". (The iTAK Package needs none of this.) With remote access set up, ATAK QR Setup also adds the Tailscale server; on iPhone pick **Phone is on: Tailscale** in the iTAK QR window for cellular.

3. The server connection turns green in the app.

Channels

Channels (TAK groups) split who sees what. Dashboard → **Channels**: add, rename or delete channels, then pick a user and tick the channels they **send** into and **receive** from. Users in no channel use `__ANON__` (everyone without a channel). Phones pick up changes when they reconnect, and users can

switch their channels on and off in ATAK/iTAK (TAK channels). `__ANON__`, ADS-B, AIS and Meshtastic are built in and cannot be renamed or deleted.

7. Remote access over cellular (Tailscale)

Remote access lets phones reach the server from anywhere they have internet (cellular or any Wi-Fi), without opening ports on a router. It uses **Tailscale**, a free private network: the server and the phones all sign in to the **same Tailscale account** and can then reach each other. Set it up once per server.

What you need

- A free Tailscale account (`tailscale.com`). Use one account for the server and all its phones.
- Internet on the server side: the node's Ethernet port, or the computer's normal internet.
- The Tailscale app on every phone (App Store / Google Play).

Step 1: connect the drive (once)

The drive has its own Tailscale identity. It is the same on an OpenMANET node, a Windows computer or a Mac, so phones set up once keep working wherever the drive runs. The computer does not need the Tailscale app.

1. Make sure the server has internet: the node's Ethernet port (or a phone tethered by USB), or the computer's normal internet.
2. Dashboard → **Remote Access** → **Enable remote access**.
3. Open the link that appears, sign in to Tailscale and press **Connect** to approve the server.
4. Back in the dashboard press **Finish setup**. The server restarts (about 2 minutes) and shows its remote address (100.x.y.z).

Step 2: set up each phone

1. Install the **Tailscale** app, sign in with the **same account**, and switch it on (allow the VPN profile when asked).
2. Set up ATAK with **ATAK QR Setup** or the **ATAK Package** (Users → your user). Either one gives the phone two servers: the local one and "*server name* (Tailscale)". On iPhone import the **iTAK Remote Package** (or scan **iTAK QR Setup** with "Phone is on: Tailscale") as a second server; keep the normal iTAK one for local Wi-Fi. Phones set up before Step 1 must be set up again.
3. Test: switch the phone's Wi-Fi off. With Tailscale on, the Tailscale server turns green over cellular.

Step 3: open the dashboard from anywhere

With remote access set up you can manage the server from anywhere with internet, not only on its Wi-Fi.

1. Find the drive's Tailscale address: dashboard → **Remote Access** ("Remote clients: 100.x.y.z or portabletak-xxxxxxxx.….ts.net"). It is the same on every node and computer the drive runs on. Write it down.
2. On the phone or laptop you will use, install the **Tailscale** app, sign in to the **same account** and switch it **on**.
3. Open a browser at `https://<Tailscale address>:8444`, for example `https://100.102.110.115:8444`. The `.ts.net` name works too.

4. The browser warns that the connection is not private (PortableTAK uses its own certificate authority). Choose **Advanced** → **Proceed** (Chrome) or **Show Details** → **visit this website** (Safari).
5. Sign in with the administrator password, the same as on Wi-Fi.

> If the page does not load: the Tailscale app must be **on** (check it shows *Connected*), the server must have internet, and the address must start with `https://` and end with `:8444`. On the server's own Wi-Fi you can also use `https://10.41.250.10:8444` (node).

If remote access does not work

Symptom	Fix
Node: "Could not start" after Enable	The node has no internet. Connect its Ethernet to a router or hotspot and try again.
"Could not start" after Enable on a computer	The computer has no internet, or a firewall blocks Tailscale. Check the internet connection and try again.
Phone stays red on cellular	Tailscale app off or signed in to a different account; the phone was set up before Step 1 (set it up again: ATAK QR Setup or a new package); the server side lost internet.
Works on cellular but not on the server's Wi-Fi	Normal for iTAK if only the Remote Package is imported: also import the normal iTAK Package.

8. Live Map

Dashboard → 🗺️ **Live Map** (top of the page), or `https://<server>:8444/map`. It shows phones (in their team colours), phones on linked servers, aircraft, ships, Meshtastic radios, markers and CASEVAC requests, updated every 3 seconds. Tap anything for its details. **Show** chooses what is on the map, **Fit all** zooms to everything, and the ☾ **Dark** / ☀ **Light** button switches the map style (the choice is remembered on that device).

Online and offline maps

- **Auto map** uses the online street map when the device viewing the map has internet, and a saved offline map when it does not. **Streets** and **Satellite** can also be picked directly (they need internet).
- **Offline maps:** while online, move the map to the area you need, open **Offline maps**, give it a name, pick the detail (**Streets** is recommended) and press **Download this area**. It is saved on the drive and works with no internet at all. A town is a few MB; a large metro area at street level about 200 MB.
- An offline map only covers its saved area. Picking it from the map menu moves the map there; outside it the map says so.
- These maps are for the Live Map. ATAK keeps its own offline maps (in ATAK: Layers → Mobile → download).

9. Live aircraft and ships (Live Feeds)

Dashboard → **Live Feeds**. Set the area (**Use my location**, or type the centre and radius), tick what you want and press **Save**. Aircraft and ships appear on the Live Map within about 20 seconds.

Source	Needs
Aircraft from the internet	Internet on the server. Free community data (adsb.lol, with adsb.fi as a fallback).
Aircraft from a USB radio	An RTL-SDR stick with a 1090 MHz antenna plugged into the node or computer. No internet needed.
Ships from the internet	A free key from aisstream.io (or an AISHub username). Ships only show near water.

To see them in ATAK, give the user **Receive** on the **ADS-B** or **AIS** channel (Channels card). Phones pick it up when they reconnect.

10. Video feeds

Live video for every TAK user. Each feed has a name; TAK clients see it in their video list, and the dashboard can play it. Feeds need no sign-in to send or watch, so anyone who can reach the server (its Wi-Fi or mesh, or your Tailscale) can watch them. Dashboard → **Video Feeds**, then pick the type when adding a feed:

Type	For
Drone / app sends RTSP	Drones such as the Skydio X10D and apps that push RTSP to the server.
TAK ICU	The TAK ICU app on a phone.
Camera / stream URL	Wi-Fi/IP cameras and other streams the server pulls itself (RTSP, RTMP, SRT or an HLS link).

Watching

- **Dashboard:** on a computer, every live feed plays on the Video Feeds page itself (it stops when you leave the page). On any screen, **Watch** plays a feed in a large window. Video is normally less than a second behind; if the fast connection is not possible the player switches to a backup and says *a few seconds behind*.
- **ATAK / WinTAK:** open **Video** (Video Player), load the list from the server and pick the feed. A phone connected over Tailscale gets the feed at the server's Tailscale address automatically. Or add it by hand: RTSP, `<server>:8554/<feed name>`, no user name or password.

Drones (Skydio X10D and other RTSP drones)

Connect the drone to PortableTAK (its position on the TAK map)

1. Dashboard → **Users** → create a user for the aircraft (e.g. `skydio1`) → **Device TLS Package**. Unzip it onto a USB-C stick. It holds the trust store, the client certificate, `secure.pref` and a **README.txt** with every value to type (the same layout as a Sit(x) TLS package). Keep it private: it contains the certificate passwords.
2. On the controller: Global Settings → Sharing → TAK/CoT → TAK Connections → Add a Connection. Server name (one word), server IP and port **8089** from the README. Advanced Options → SSL/TLS:

the trust store (`truststore-root.p12`) and the client certificate (`skydio1.p12`), each with the password from the README.

Step 1: add a feed (dashboard)

1. Dashboard → **Video Feeds** → type a feed name, keep the type **Drone / app sends RTSP** → **Add feed**. Use lowercase letters, numbers and dashes, for example `skydio1`. One name per aircraft.
2. The card then shows the exact drone settings, with this server's address. **Drone settings** shows them again later.

Step 2: set up the Skydio X10D controller

1. Connect the controller's Wi-Fi to the same network as the server (the node's Wi-Fi, or the Wi-Fi the Windows/Mac computer uses).
2. Menu (three lines, top left) → **Sharing** → **RTSP** → **External**.
3. Stream name: the feed name (e.g. `skydio1`). IP address: the server's address shown in the card (node: `10.41.250.10`). Port: **8554**.
4. **Start Stream**. The video starts once the aircraft takes off; the card shows **Live**.

Step 3: watch it

ATAK / WinTAK: open **Video** and pick the feed by name. Or add it by hand: type RTSP, address `<server>:8554/skydio1`, and leave user name and password blank.

Flying away from the server: Skydio on a phone hotspot

When the drone team is on cellular and the server is elsewhere, the controller joins an ATAK phone's hotspot and that phone forwards the drone's traffic to the server over its own Tailscale. Nothing is installed on the controller; the phone needs the free **Termux** app. Dashboard → **Video Feeds** → **Drone far from this server** has the one command to paste. Step by step: **PortableTAK Skydio X10D Guide** (Setup B).

TAK ICU (phone camera)

1. Dashboard → **Video Feeds** → name the feed (e.g. `icu`), type **TAK ICU** → **Add feed**. **TAK ICU settings** then shows the exact values for this server.
2. In TAK ICU: menu → Settings → **Broadcast Preferences**: **Destination Type** *Local Area Network*, **Delivery Method** *RTSP*, **RTSP Server Address** `<server>:8554/_in/<feed name>` (node: `10.41.250.10:8554/_in/icu`). Under Media Preferences, **Medium (480p)** is a good resolution for Wi-Fi and Tailscale.
3. Press the red button to broadcast. The feed shows **Live**.

The server receives TAK ICU's video over a reliable connection and repackages it so browsers can play it, so moving the camera does not smear the picture. **Watching on the same phone** that runs TAK ICU only works in split screen (both apps visible): Android stops the camera of an app in the background. ATAK may also list a second feed that TAK ICU announces itself (UDP); it does not play, delete it and use the one from the server.

> While any TAK ICU feed exists, the server accepts RTSP video only over TCP (TAK ICU switches to it by itself; RTSP over UDP loses picture over Wi-Fi). A sender that cannot use TCP cannot push while a TAK ICU feed exists.

Cameras and other streams

1. Dashboard → **Video Feeds** → name the feed, type **Camera / stream URL**, paste the camera's address, for example `rtsp://user:password@192.168.1.20:554/stream1` (see the camera's manual), → **Add feed**.
2. The server connects to the camera itself, all the time, and the feed shows **Live** once it has video. The camera must be reachable from the server (same network, or Tailscale).

The password stays on the drive and is shown as `***`. ATAK plays H.264 and H.265; browsers need H.264 on most computers (most cameras can be set to H.264).

11. ATAK plugins and data packages

Dashboard → **ATAK Plugins** → **Upload plugin (APK)**. Phones set up with this server's QR code or package find the plugins in ATAK under **Plugins** (⋮ → **Sync**) and install them without internet. **On setup** installs a plugin when a phone enrolls; **On connect** offers it every time a phone connects. Upload a newer APK of the same plugin to update it.

Data packages

Dashboard → **Data Packages**: the server's data packages, the ones phones send and share and ones uploaded here. **Upload** a ready-made ATAK data package (.zip) or any file (map overlay KML/KMZ, PDF, image, document); other files are wrapped in a data package automatically. **Send** picks connected phones: ATAK shows its "file received" notice, downloads it from this server (over Wi-Fi, mesh or Tailscale) and opens it (KML/KMZ appear on the map). **Download** saves a copy; **Delete** removes it from the server (phones keep theirs). Up to 256 MB per file.

12. Meshtastic

Links Meshtastic radios to the server: radios appear on the Live Map and, for users who receive the **Meshtastic** channel, in ATAK; text messages pass between Meshtastic and ATAK chat. A radio connects over Wi-Fi (radios with Wi-Fi on the server's network) or through the Meshtastic phone app (MQTT client proxy).

1. Dashboard → **Meshtastic** → **Turn Meshtastic on** → **Apply now** (the server restarts, about 2 minutes).
2. In the Meshtastic app for each radio: Settings → Modules → **MQTT**: enter the server, username, password and root topic the card shows; encryption on, JSON off, TLS off. Then for each channel: Channels → channel → MQTT: **Uplink** and **Downlink** on. A radio without Wi-Fi: also turn on **MQTT client proxy** (the phone relays).
3. Private channels: in the card, **Add channel** with the channel link from the Meshtastic app (Share). Applies after a restart.

A Meshtastic radio plugged into a node by USB (for example one used by the ManetNode field extension) is not used by this feature; it connects over MQTT only.

13. Voice (push to talk)

The server runs a voice server (Mumble, port 64738) for push to talk. It starts with PortableTAK; there is nothing to set up. Phones use the **TAK PTT** ATAK plugin, which signs in with the same certificate the phone uses for TAK, so there are no extra accounts or passwords. Other Mumble apps (such as Vx) sign in with a PortableTAK user name and password. Turning a user off or deleting them on the dashboard also locks them out of voice.

1. Install the **TAK PTT** plugin in ATAK (dashboard → **ATAK Plugins**, or the plugin file) and load it.
2. Connect ATAK to this server as usual. TAK PTT finds the voice server and signs in by itself (the pill at the top shows **CONNECTED**; tap it to see the server and your user name).
3. Allow the microphone when asked. **PTT** tab: hold the ring to talk (or set tap to start, tap to stop under **SET**). A short beep means you may talk; talking stops after the time limit (60 seconds, 20 to 90 under SET).
4. **CHANNELS** tab: tap a channel's name to move there, **+ NEW CHANNEL** to make one. **RX** listens to a channel without leaving yours; **TX** picks the channels you talk on (several at once).
5. **HISTORY** tab: transmissions from the last 7 days, newest first, with who, when and from where. Tap one to play it; **PLAY UNHEARD** plays the new ones in order; **MAP** shows where each was sent from (on your map only). **TEXT** shows what was said under each one (see Transcripts below). Live traffic always pauses playback.

SOS (the red circle beside the PTT ring): hold it for 3 seconds to send ATAK's 911 emergency alert to everyone; tap it again to cancel. When the voice server cannot be reached, TAK PTT keeps working over the local network (OpenMANET or Wi-Fi) between phones that have connected to this server before; the pill shows **MESH**. It goes back to the server by itself.

OpenMANET talk groups: tap **CHANNELS** again to switch to **OPENMANET** (cyan): talk groups CH 1 to CH 5, the same ones OpenMANET nodes and the ManetNode plugin use, each with RX and TX. They work without the server too. The server also joins its **LOBBY** to talk group 1 (the user **OpenMANET CH 1** in LOBBY), so people on the server and people only on the mesh hear each other. To link other channels, set `"talkGroups": "LOBBY=1,Ops=2"` in `opentak/voice.json` on the drive (`""` turns the link off), then restart PortableTAK. One person talks at a time across the link, like a radio. Talk groups are not encrypted: anyone on the mesh can listen. The link needs the server on the mesh's network (an OpenMANET node or a Linux computer on it); on a Mac or Windows computer, where PortableTAK runs in a virtual machine, talk groups may not get through.

Transcripts: with **TEXT ON** in HISTORY, the phone turns each recorded transmission into text and shows it under the clip. OpenMANET talk-group transmissions the phone hears are turned into text too: the newest shows on the PTT tab's status line (in cyan) for 20 seconds, and the last ones are listed at the top of HISTORY under **HEARD ON THIS PHONE**, also without the server. It all happens on the phone, without internet, using Android's on-device speech recognition (Android 13 or later; on phones without it, TEXT says so). Nothing is sent anywhere. Clear speech works best; noise, radio static and several people at once give gaps or wrong words.

Talking with ATAK in the background: under **SET**, turn on the **floating PTT button** (Android asks to let TAK PTT appear on top) or **volume key as PTT** (Android's accessibility settings: turn on **TAK PTT buttons**; it only watches that key). While voice is not connected the volume key works normally.

Ports: the launchers on Mac, Windows and Linux open 64738 (TCP and UDP). Voice history is served on 8443 at `/voice/` to phones with a PortableTAK certificate only. Clips are kept 7 days (and at most 2 GB).

14. Linking servers (federation)

Federation shares positions, markers, alerts and chat between two TAK servers, so each team sees the other on its map. It works with another PortableTAK, OpenTAKServer with PortableTAK's federation add-on, and the official TAK Server (tak.gov). Dashboard → **Federation** (under Remote Access).

Data packages an ATAK user sends to someone on the other server also cross between PortableTAK servers: the file moves over the federation link and the receiving phone downloads it from its own server. Not yet with an official TAK Server, and Data Sync missions are not shared.

Two PortableTAK servers: one code

1. On the first server: **Invite a server**. Copy the code (or download the invite file) and send it to the other server's admin. It works once, for 24 hours.
2. On the second server: **Join a server**, paste the code, **Join**. Both servers trust each other, share with everyone, and the link turns green within a minute.

The servers need a network path to each other: the same network, or both drives' Remote Access signed in to the same Tailscale account.

An official TAK Server

1. **Connect to a TAK Server**: name, address and federation port (9001), then **Check**. PortableTAK reads the TAK Server's certificate authority itself and shows its fingerprint; have the TAK Server admin confirm it if you can, then **Trust and connect**.
2. Send the TAK Server admin the sheet that appears (it is also under Advanced): this server's CA certificate to upload under *Federation > Manage Federate Certificate Authorities*, then, once PortableTAK has connected, the groups to allow for its federate (users without a group are `__ANON__`).

Choosing what flows

Each linked server shows this server, two arrows and the other server. Tap → to send to it or stop sending, ← to receive from it or stop receiving (green = on); the line underneath says which way data flows. **Pause** stops a link without deleting it. **Channels...** on a link chooses which channels it sends and which channels the other server's data arrives in (official TAK Servers match channels by name).

Good to know

- A server that connects to yours without an invite appears as *not sharing yet*: press **Allow sharing** only if you trust it.
- Each link is locked to the other server's certificate. If that server is rebuilt, use **Forget its certificate** (or remove the link and invite again).
- Moving the drive to another node or computer keeps its links: the federation certificate belongs to the drive.

15. Backups, moving and erasing

- **Backups: Backup / Restore** creates checked backups on the drive. Download important ones and keep them elsewhere.
- **Moving the drive** to another node or computer: stop or safely remove it, plug it in elsewhere. It adapts by itself.
- **Erase Drive** (bottom of the dashboard): returns the drive to brand new. It deletes all users, certificates, settings, remote access, data and on-drive backups. It asks for the administrator password and the word ERASE, and cannot be undone. Set the drive up again with PortableTAK Setup afterwards.

16. Troubleshooting

Problem	What to do
Phone says "host unreachable" or never connects	Check the phone is on the node's Wi-Fi (or the same network as the Windows computer). Use the right address (node: 10.41.250.10). On Android, set the Wi-Fi to DHCP (not static); if it still fails, forget and rejoin the network. Over cellular, the Tailscale app must be on.
Connection red only on cellular	Remote Access must be enabled and finished, the node/computer needs internet, and the phone's Tailscale app must be connected.
Another device shows grey or disappears	Make sure each device has a different callsign. Grey can also mean that device lost its connection; it returns when it reconnects.
iPhone sees only one server	Expected: iTAK reads one server per package. Import the **iTAK Remote Package** as well.
ATAK says "registration failed" or "TAK Server profile request failed" after QR setup	The phone scanned a QR made before an update, or the server's address changed since. Delete the server in ATAK, then scan a fresh **ATAK QR Setup**. If the server still turns green, it is harmless.
iTAK QR: "There was an error connecting to the server"	The iPhone does not fully trust this drive yet. Installing the **iPhone Trust Profile** is not enough: also go to Settings → General → About → **Certificate Trust Settings** and switch **on** this server's certificate. Each drive has its own profile (one made for another drive does not count). The **iTAK Package** works without this.
TAK PTT says OFFLINE or "Connect ATAK to your PortableTAK server first"	ATAK must be connected to this server (green) first; TAK PTT uses that connection and its certificate. Then tap the pill and **Reconnect**. Over the network, port 64738 must be reachable (the launchers open it).
TAK PTT: "Voice sign-in refused"	The user is turned off or was deleted on the dashboard, or the phone's certificate is from another server. Turn the user back on, or re-enroll the phone with this server.
TAK PTT: no sound or nothing sent with ATAK in the background	Allow the microphone for TAK PTT (tap the status line). For the floating button, allow **Appear on top**; for the volume key, turn on **TAK PTT buttons** in Android's accessibility settings.
"Invalid credentials" / QR enrollment fails	Use the user's own name and password, not the administrator. ATAK must trust the server first (import the ATAK Trust Package once). Generate a new QR (each works 3 times).
Certificate warning in the browser	Normal: the server uses its own certificate authority. Choose Advanced → Continue. To remove it, install the drive's trust certificate on that device.
Dashboard sign-in refused after several tries	Wait a few minutes (sign-in is rate-limited per device), then use the administrator password chosen at setup.
Node: server does not start after plugging in	Wait 5 minutes. Re-seat the drive firmly in a blue USB 3 port. Power-cycle the node; PortableTAK starts on boot.

Windows: "No PortableTAK drive found"	Plug the drive in and wait until the PORTABLETAK drive appears in File Explorer, then start again.
Windows: asks to restart	WSL was just installed. Restart, then start PortableTAK again.
Windows: phones cannot connect	Phones must be on the same network as the computer. Changed Wi-Fi since starting? Use **Stop PortableTAK** and start it again so it uses the new address, then set phones up again (QR or package). Windows 11 22H2+ is required.
ATAK: "The TAK Server's identity could not be verified" after QR setup	ATAK no longer trusts the drive (deleting servers in ATAK can remove the trust). Import the **ATAK Trust Package (for QR)** again from Client Setup, then scan a fresh QR. Or import the **ATAK Package** for that user, which carries everything ATAK needs.
Federation link stays red	"Not connected: certificate" after the other server was rebuilt: **Forget its certificate**, or remove and invite again. "Connection refused" or unreachable: the servers need a network path (same network, or the same Tailscale account). A TAK Server peer: its admin must upload this server's CA and set the federate's groups.
Drone feed never shows Live	The controller must be on the server's network, with the IP address and port **8554** from the Video Feeds card and exactly the same stream name. Video only starts once the aircraft is flying.
Linux: "no PortableTAK drive found"	Plug the drive in and wait a few seconds, then `sudo portabletak start`. Check it shows in `lsblk`.
Linux: Live Feeds says no USB radio, or the radio sends no data	Unplug the stick and plug it in again, then `sudo portabletak start` (it releases the stick from the TV-tuner drivers). If it still fails, restart the computer once.
Linux / Raspberry Pi: "needs a 64-bit computer"	Install the 64-bit version of the system (Raspberry Pi OS 64-bit). 32-bit systems and Pis older than the Pi 4 cannot run PortableTAK.
Windows: drive vanished from File Explorer	Normal while PortableTAK runs. Use **Stop PortableTAK** to get it back.
Windows offers to format the drive, or macOS to initialize it	Always Cancel / Ignore. The Linux partition holds the server.
"More than one PortableTAK drive is attached" / server does not start with two drives plugged in	Use one PortableTAK drive per node or computer at a time. Unplug the other drive and start again (node: power-cycle or re-plug).
The drive was unplugged without Safe Remove / Stop PortableTAK (or power was lost)	Usually no harm: plug it back in and start PortableTAK. It checks and repairs the drive automatically and cleans up the previous session. Next time use **Safe Remove** (dashboard) or **Stop PortableTAK** (desktop) first.
Live Map is blank	No internet on the viewing device and no offline map for that area, or you are looking outside a saved offline map: pick it in the map menu (it moves there), or save the area while online.

No aircraft on the Live Map	Internet aircraft need internet on the server and an area set in Live Feeds. The card shows the last error. A USB radio must be plugged in before saving (the card says "USB radio detected").
Watch says "a few seconds behind"	The fast (WebRTC) connection could not reach the server; the backup player is used. It still works; check the browser can reach the server directly (same network or Tailscale).
TAK ICU video smears or freezes	Use the settings from the TAK ICU settings section (Local Area Network, RTSP, `…:8554/_in/<feed>`). Frozen on the same phone: use split screen. Delete ATAK's extra UDP entry for TAK ICU.
Camera feed stays "Connecting to camera"	Check the URL, user name and password (try it in VLC on a computer on the same network), and that the server can reach the camera.
Plugin does not show in ATAK	The phone must be set up with this server's QR code or package. In ATAK: Plugins → ⋮ → Sync.
Meshtastic radio not on the map	Meshtastic must be on and applied; the radio's MQTT settings must match the card exactly, with Uplink on for its channel. A private channel must be added to the card (then restart).
Map icons missing	They download automatically the first time the server runs with internet.
Forgot the administrator password	It cannot be recovered. Use a backup made elsewhere, or erase and set up the drive again.
Node: a new node does not start PortableTAK	The node needs the PortableTAK loader once (field extension or `portabletak-loader-install.sh`). On the node, `portabletak-loader status` shows what happened; a refused drive means its package is not signed with the PortableTAK release key (make the drive again with the official Setup).

Logs

Linux: `journalctl -u portabletak` and `/opt/portabletak/root/tmp/portabletak/`. **Windows:** `%LOCALAPPDATA%\PortableTAK\portabletak.log`. **Node** (administrators, over SSH): `/usr/local/portabletak/bin/portabletak-health` and the logs under `/mnt/asgard/opentak/data/logs`.

17. Security notes

- The dashboard requires sign-in. It is reachable on the TAK network at port 8444 (HTTPS), and with remote access also from devices signed in to your Tailscale account (nobody else can reach it). It is protected against cross-site requests and password guessing. Choose a strong administrator password for real use.
- First-time setup can only be done on the computer that runs the drive, never from a phone.
- The drive contains the server's private keys. Keep it physically secure. Erase drives before giving them away.